

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	8.9/10
2021	XII(CBSE)	Government Model Senior Secondary School, Chandigarh	96%
2019	X(CBSE)	DAV Public School, Ambala	97.8%

SCHOLASTIC ACHIEVEMENTS

- Received **Academic Excellence Award** for outstanding academic performance (top 10%) by IIT Kanpur. (Dec'22)
- All India Rank **1086** in Joint Entrance Examination Advanced among 0.2 million candidates all over the nation. (Oct'21)
- All India Rank **3038** in Joint Entrance Examination Mains among 1.2 million candidates all over the nation. (Sep'21)
- State Topper** of Chandigarh in Indian National Physics Olympiad (IOQP) among total 250 candidates in state. (Mar'21)
- Among **Top 347** shortlisted candidates from **20,858** all over the nation in Indian National Physics Olympiad (IOQP). (Mar'21)
- Achieved **Gold Level** in **World Quant Challenge**, awarded with Certificate of Accomplishment. (June'23)

WORK EXPERIENCE

- Determining Classroom Engagement | SURGE'23 | Faculty Advisor: Dr.Tushar Sandhan   (May'23 - July'23)
- Co-Authored a paper on **Classroom engagement evaluation using 360-degree view of the camera with deep learning techniques** in the **International Society for Optics and Photonics (SPIE)** conference 2023.
 - Preprocessed and annotated raw dataset of back camera images obtained from Classroom environment using **Roboflow Api**.
 - Trained dataset using **Custom YOLOv5 object detection algorithm** to analyse **Student's gaze** towards projector or not.
 - Collected and preprocessed audio dataset for training, extracted **MFCC features** from dataset which is used as training dataset.
 - Implemented **1D CNN network** using Keras Sequential API for classifying audio into "Engaging" and "Sleep inducing" classes.

KEY PROJECTS

- Business Analytics using Python** | Finance and Analytics Club, IIT Kanpur, IITK (May'22 - July'22)
- Learned to apply **Data analytics** tools and techniques like **Python Pandas** to improve upon strategic **Business decisions**.
 - Studied **Data Handling and Manipulation techniques** like **filtering, sorting, grouping** etc. are applied over sample datasets.
 - Learned various **Data Mining** techniques like **Decision Tree, Cluster Analysis and Association rule**.
- Applications of Reinforcement Learning** | Association for Computing Activities (ACA) , IITK  (May'22 - July'22)
- Introduced to **Markov Decision Process, Policy, Rewards and Actions** to understand basics of reinforcement learning.
 - Learned about **States Transition Graph, Value Iteration and Policy Iteration** methods to Optimally evaluate Policy.
 - Completed Programming assignments on **Markov decision process and Optimum Policy Evaluation** in various environments.
- Building Own Computer Vision Library** | Stamatics, IITK (May'23 - July'23)
- Acquainted to basics of Pixels, ways of storing images in form of matrix and different types of colorspace (RGB and HSV).
 - Introduced to **Image resizing, interpolation (various types of interpolations), filtering and convolution** techniques.
 - Implemented various Custom Image Manipulation functions towards creating our own library as a programming assignments.
- Algo-Trading** | Finance and Analytics Club, IITK  (May'22 - July'22)
- Studied basics of **Stock market, Options and Crypto-Currencies** and devised strategy for **Intraday Trading**.
 - Derived **Probabilistic weights** and performed analysis to sort the list of best stocks for an **Intraday** strategy.
 - Tried to perform **Time Optimizations** to find the best time to enter and leave market for high returns using Python.

TECHNICAL SKILLS

Programming Languages	C, C++, Python, HTML, JavaScript
Tools & Libraries	Numpy, Git, Fusion 360, Seaborn, Pandas, Numpy, Matplotlib, YOLOv5
Miscellaneous	CSS3, L ^A T _E X, AutoCAD, Colab, Jupyter notebook, Markdown

RELEVANT COURSES

* Online

Data Structures and Algorithms	Fundamentals of Computing	Linear Algebra and ODE
Complex Analysis	Real Analysis	Probability and Statistics
Introduction to Electronics	Microelectronics	Signals, Systems and Networks
Partial Differential Equations	Neural networks and Deep Learning*	Improving Deep Neural Networks and Optimizations*

PROJECTS MENTORED

- Mentor, Coding Ninjas (Dec'22 - Apr'23)
- Helped students by resolving doubts, debugging codes related to question on **Data Structures and Algorithms**.
 - Assisted **50+** students in gaining complete knowledge of subject, helped them in their assignments and projects.
- Supervised Machine Learning for Business analytics, ACA, IIT Kanpur  (May'23 - July'23)
- Mentored this project to a group of **20+ individuals** to get introduction to Machine learning and it's application in Business.
 - Introduced to all types of **Regression models, Data Handling, Decision trees** and various other techniques.

POSITIONS OF RESPONSIBILITY

- Executive, Stamatics, IIT Kanpur (Oct'22 - Present)
- Organised Mathemania and Integration-Bee events for **freshers batch (2022)** of **1100** students with other 20 team members.
- Academic Mentor, Counselling service, IIT Kanpur (Sep'22 - Present)
- Conducted remedial classes at Institute Level and Hall Level for around **100 students** for the course PHY112 and PHY115.